Computational Imaging

BMED365: Topic List Y01–Y10

BMED365

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Overview

1. **Fundamentals of Computational Imaging**
   - Y01: Fundamental concepts in computational imaging
   - Y02: The imaging pipeline
   - Y03: Digital image representation
   - Y04: Basic image processing operations
2. **Advanced Imaging Techniques**
   - Y05: Imaging Mass Cytometry (IMC)
   - Y06: Multiplexed tissue analysis
3. **AI in Medical Imaging**
   - Y07: Role of AI in medical image analysis
   - Y08: Image segmentation approaches
   - Y09: MedSAM (Segment Anything in Medical Images)
   - Y10: LLMs integrated with medical imaging

Y01: Explain fundamental concepts in computational imaging

Computational imaging = **combining optics, sensors, and computation**

Definition

Computational imaging refers to imaging techniques where the final image is **computed** from raw sensor data, rather than directly captured.

Key concepts:

- **Image formation:** How physical signals become digital data
- **Reconstruction:** Algorithms that transform raw data into images
- **Enhancement:** Improving image quality computationally
- **Analysis:** Extracting information from images

Medical relevance

CT, MRI, PET, and ultrasound all rely on computational methods to produce diagnostic images from raw sensor data.

Y02: Describe the imaging pipeline: acquisition, processing, analysis

The imaging pipeline:



Acquisition:

- Scanner/sensor captures raw data
- Parameters: resolution, contrast, noise

Pre-processing:

- Noise reduction, artifact correction
- Format conversion, normalization

Processing:

- Filtering, enhancement, registration
- Feature extraction

Analysis & Visualization:

- Segmentation, quantification
- 2D/3D rendering, reporting

Y03: Understand digital image representation

Digital images = discrete samples of continuous signals

Pixels (2D):

- Picture elements
- Arranged in a grid (rows $\times$ columns)
- Each pixel has an intensity value

Voxels (3D):

- Volume elements
- Extension of pixels to 3D
- Used in CT, MRI volumes

Intensity representation:

- **8-bit:** 0–255 (256 levels)
- **12-bit:** 0–4095 (medical imaging)
- **16-bit:** 0–65535
- **Floating point:** continuous values

Image types:

- Grayscale (single channel)
- RGB color (3 channels)
- Multi-channel (hyperspectral, IMC)

Key parameters

Resolution: Number of pixels/voxels. **Bit depth:** Number of intensity levels. **Spacing:** Physical size of each pixel/voxel.

Y04: Explain basic image processing operations

Image processing = **manipulating images to improve quality or extract information**

Filtering operations:

- **Gaussian blur:** Smoothing, noise reduction
- **Median filter:** Remove salt-and-pepper noise
- **Sharpening:** Enhance edges
- **Edge detection:** Find boundaries
- **Morphological ops:** Erosion, dilation

Enhancement techniques:

- **Histogram equalization:** Improve contrast distribution
- **CLAHE:** Adaptive local contrast enhancement
- **Intensity normalization:** Standardize intensity ranges

In medical imaging

Careful processing is essential – over-enhancement can create artifacts or hide pathology. Always preserve diagnostic information.

Y05: Describe Imaging Mass Cytometry (IMC) and its applications

Imaging Mass Cytometry (IMC):

Definition

IMC combines mass spectrometry with imaging to simultaneously detect **40+ protein markers** in tissue sections at subcellular resolution.

How it works:

1. Tissue is stained with **metal-conjugated antibodies**
2. Laser ablates small spots (1 μm resolution)
3. Mass spectrometer identifies metals $\rightarrow$ protein markers
4. Computational reconstruction creates multiplexed image

Applications in biomedicine:

- **Tumor microenvironment:** Map immune cell infiltration
- **Cell phenotyping:** Identify cell types and states
- **Spatial biology:** Understand tissue architecture

Y06: Explain how IMC enables multiplexed tissue analysis

Multiplexed analysis = many markers simultaneously

Advantages over traditional immunohistochemistry:

Traditional IHC:

- Limited to 2–4 markers
- Multiple serial sections needed
- Spectral overlap limits colors

IMC:

- 40+ markers on single section
- No spectral overlap (mass tags)
- Preserves spatial relationships

Computational analysis pipeline:

1. **Cell segmentation:** Identify individual cells
2. **Feature extraction:** Measure marker intensities per cell
3. **Clustering:** Group cells by phenotype
4. **Spatial analysis:** Analyze cell neighborhoods and interactions

Key insight

IMC generates **high-dimensional spatial data** requiring sophisticated computational methods for analysis.

Y07: Understand the role of AI in medical image analysis

AI is transforming medical imaging:

Key applications:

- **Detection:** Find abnormalities
- **Segmentation:** Delineate structures
- **Classification:** Diagnose conditions
- **Quantification:** Measure volumes, changes
- **Prediction:** Prognosis, treatment response
- **Image enhancement:** Super-resolution, denoising

Why deep learning excels:

- Learns features automatically from data (no hand-crafted features)
- Handles high-dimensional image data naturally
- CNNs designed for spatial data

Clinical deployment considerations

AI tools require regulatory approval (FDA, CE marking), validation on local populations, and integration into clinical workflows.

Y08: Describe image segmentation approaches

Image segmentation = **partitioning image into meaningful regions**

Classical approaches:

- **Thresholding:** Intensity-based separation
- **Region growing:** Expand from seed points
- **Watershed:** Topographic flooding
- **Active contours:** Deformable models

Pros: Interpretable, no training data needed
Cons: Sensitive to noise, requires tuning

Deep learning approaches:

- **U-Net:** Encoder-decoder with skip connections
- **nnU-Net:** Self-configuring U-Net
- **Vision Transformers:** Attention-based
- **SAM/MedSAM:** Foundation models

Pros: State-of-the-art accuracy, handles complexity
Cons: Requires training data, less interpretable

Y09: Know about MedSAM (Segment Anything in Medical Images)

MedSAM = Medical adaptation of Segment Anything Model

Background:

- SAM (Meta AI, 2023): Foundation model for image segmentation
- Trained on 11M images, 1B masks – generalizes to new tasks
- **MedSAM:** Fine-tuned on 1.5M medical image-mask pairs

Key features:

- **Promptable:** Specify what to segment with points, boxes, or text
- **Zero-shot:** Works on unseen image types
- **Interactive:** Refine with additional prompts
- **Multi-modality:** CT, MRI, X-ray, ultrasound, microscopy
- **Efficient:** Fast inference

Impact

MedSAM represents a shift toward **foundation models** in medical imaging – general-purpose models that can be adapted to specific tasks with minimal effort.

Y10: Discuss how LLMs are integrated with medical imaging

Large Language Models (LLMs) + Medical Imaging:

Integration approaches:

1. **Vision-Language Models:** Joint understanding of images and text
2. **Report generation:** Automatically describe findings
3. **Question answering:** "What abnormalities are visible?"
4. **Clinical decision support:** Combine image analysis with patient context

Examples:

- **Med-PaLM M:** Google's multimodal medical AI
- **LLaVA-Med:** Open-source medical VLM
- **GPT-4V/Claude with vision:** Can analyze medical images (with caveats)

Current limitations

Hallucinations: LLMs may generate plausible but incorrect findings. **Not FDA-approved** for clinical diagnosis. Use as assistive tools only, with human oversight.

Summary: Computational Imaging

Key points:

- **Y01:** Computational imaging combines optics, sensors, and computation
- **Y02:** Imaging pipeline: acquisition $\rightarrow$ processing $\rightarrow$ analysis
- **Y03:** Digital images: pixels/voxels with intensity values
- **Y04:** Image processing: filtering, enhancement, feature extraction
- **Y05:** IMC enables 40+ marker detection in tissue
- **Y06:** Multiplexed analysis reveals cell phenotypes and spatial patterns
- **Y07:** AI excels at detection, segmentation, and classification
- **Y08:** Segmentation: classical methods vs. deep learning (U-Net)
- **Y09:** MedSAM: foundation model for medical image segmentation
- **Y10:** LLMs + imaging: report generation, VQA, multimodal analysis

Key message

Computational imaging is essential for modern medical diagnostics. AI and foundation models are rapidly advancing capabilities, but require careful validation and human oversight in clinical settings.